

What the loss hid

The task coordinate was written off on the strength of a denoising loss that barely moves. In generated samples it decides the outcome every time.

cifar_ds3 @ 50k · validation · 24 episodes · 128 samples per cell · DDIM $\eta = 0$, 50 steps

100 %

of generated sets carry the **right transformation** given the episode's own coordinate — against 33 % chance

1.7 %

is all the **same coordinate** is worth on the denoising loss the project has been reading

5.7×

closer to the target domain than reusing the source coordinate, at $K_T = 1$

The picture without any instrument

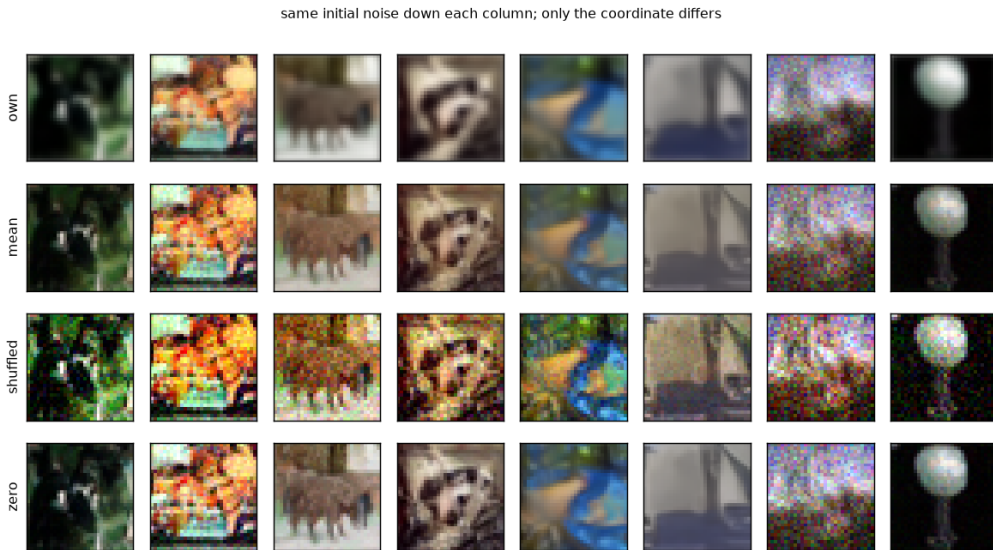


Figure 1: Each column starts from **one shared initial noise**, so composition is fixed down the column and only the coordinate changes. Columns 1–6 are blur tasks, 7–8 noise tasks. The **own** row follows the task; **mean** and **zero** ignore it; **shuffled** follows the *wrong* task.

Deployment strategies, re-run in this metric

Distance from the real target domain, lower is better. The denoising loss puts all six rows on the same number.

strategy	$K_T = 1$	$K_T = 5$	$K_T = 20$
no adaptation	1.577	1.614	1.607
reuse z_S	1.410	0.805	0.484
transport	0.249	0.272	0.271
transport + refine	0.245	0.300	0.356
target only	0.276	0.285	0.253
oracle	0.258	0.253	0.249

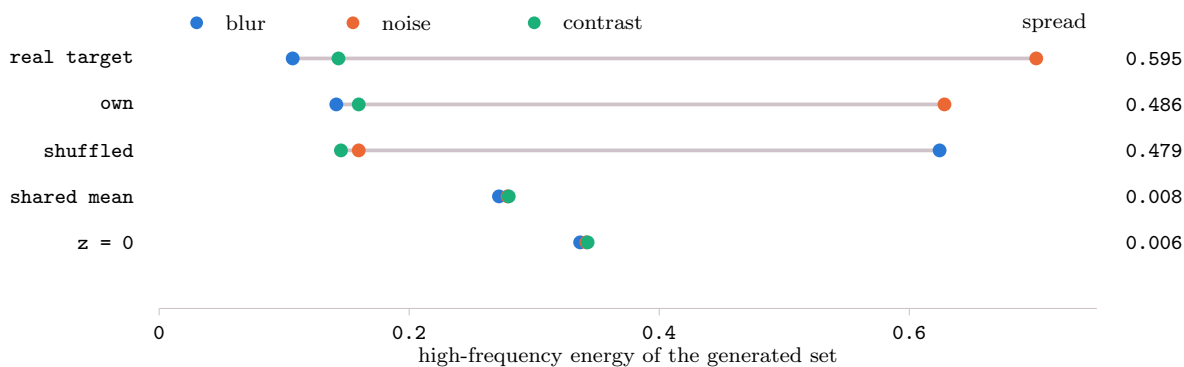


Figure 2: Blur suppresses high-frequency energy, added noise raises it. A coordinate that carries the task spreads its three values apart. **own** reproduces 82 % of the real spread; **mean** and **zero** collapse onto a point; **shuffled** spreads just as far, in the wrong direction — which is what rules out every explanation that does not run through the coordinate.

- **Transport beats reusing the source coordinate** by $+1.161 \pm 0.233$ at $K_T = 1$, shrinking to $+0.213$ at $K_T = 20$ — the shape the method predicts, and the shape stage 1 produced on Gaussian mixtures.
- **Transport does not beat encoding one target image** ($+0.027 \pm 0.096$). The transformation is too easy to read from a single sample for source information to add anything.
- **Refinement makes samples worse, and worse with more data** ($0.245 \rightarrow 0.300 \rightarrow 0.356$). It optimises the loss, and the loss is the thing that disagrees with the samples.

What this changes

- **The representation was never the problem.** The instrument that said so compresses the effect by roughly fifty-fold. The earlier “0 % task-specific” result reproduces exactly on the arm it was measured on — one transformation, 32 channels — and does not generalise to the arm where tasks differ in something denoising depends on.
- **Switch refinement off** in this setting, or re-point it at an objective that agrees with the samples.
- **The centring change is no longer the indicated fix.** It was the remedy for a representation with no task-specific content, and that diagnosis has not survived.

What these numbers do not say

- Nothing clean about **semantic class**: the coordinate carries transformation identity ($+0.00133 \pm 0.00035$, 98.5 % of the effect) and almost no class identity ($+0.00002 \pm 0.00001$, 1.5 %).
- Generation reaches **82 %** of the way to the target domain, not all of it — and the oracle does no better, so the residual gap is the model’s reach rather than a mis-set coordinate.
- Everything here is the **three-transformation checkpoint**. The single-transformation arms show $\Delta_{\text{task}} \approx 0$ and have not been put through the generation test.